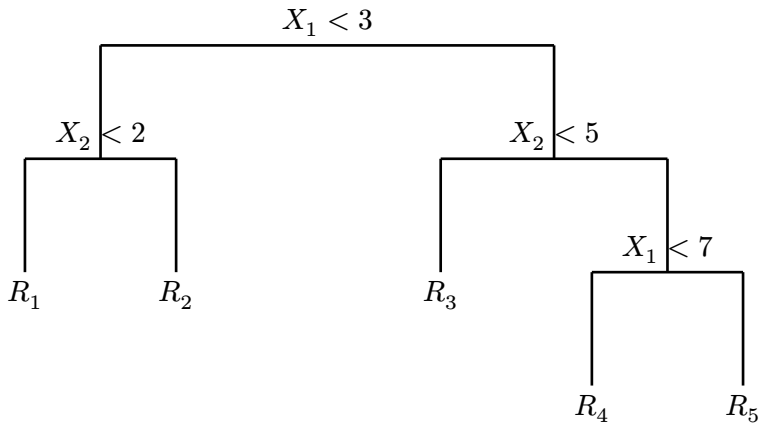
The University of Texas at Austin
In-Class Work 8

M339G Predictive Analytics

November 03, 2026

Regression Trees.

Problem 8.1. You are given the following regression tree with predictors X_1 and X_2, where the inequality at each split describes the left branch:



Which of the following correctly describes region R_4?

a. $\{X \mid X_1 < 7, X_2 \geq 5\}$
b. $\{X \mid X_1 \geq 7, X_2 \geq 5\}$
c. $\{X \mid 3 \leq X_1 < 7, X_2 < 5\}$
d. $\{X \mid 3 \leq X_1 < 7\}$
e. $\{X \mid 3 \leq X_1 < 7, X_2 \geq 5\}$

Problem 8.2. A regression tree is fit using recursive binary splitting. Which of the following statements about the resulting regions $R_1, \ldots, R_J$ is true?

a. Two regions may overlap, so an observation can belong to more than one region.
b. The number of regions equals the number of internal (splitting) nodes.
c. Each predictor can be used to split at most once in the tree.
d. The regions are non-overlapping boxes whose union is the entire predictor space.
e. A boundary between two regions may be a diagonal line such as $X_1 + X_2 = 4$.

Problem 8.3. A regression tree has been fit to a training data set. Consider the following statements:

I. The predicted response for a new observation is the mean of the training responses in the terminal node it falls into.
II. Two new observations that fall into the same terminal node receive the same predicted response, even if their predictor values differ.
III. The predicted response for a new observation can exceed the largest response in the training data.

Determine which of the statements are true.

a. I only
b. II only
c. I and II only
d. I and III only
e. I, II, and III

Problem 8.4. You are given the following training data with one predictor X and response Y:

X	1	2	3	4	5
Y	2	3	9	10	11

The first split of a regression tree is chosen by recursive binary splitting from the candidate cutpoints $s \in \{1.5, 2.5, 3.5, 4.5\}$, with the left branch $X < s$. Determine the cutpoint chosen.

(A) 1.5
(B) 2.5
(C) 3.5
(D) 4.5
(E) It cannot be determined without a validation set.

Problem 8.5. Consider the following statements about recursive binary splitting for regression trees:

I. It is a greedy algorithm: at each step it makes the best split for that step, without looking ahead to splits that might lead to a better tree later.
II. Among all trees with the same number of terminal nodes, it is guaranteed to produce the one with the smallest training RSS.
III. At each step, it considers every predictor and every possible cutpoint for that predictor.

Determine which of the statements are true.

a. I and III only
b. I only
c. II only
d. I and II only
e. I, II, and III